



Sizhang Xu

Full-Stack Engineering · Real-Time Systems

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EDUCATION

University of Washington

M.S. in Technology Innovation · Global Innovation Exchange (GIX) · In progress

Sep 2026 – Jun 2028 (expected)

Tsinghua University

Graduate studies · GIX dual-degree program

Lab affiliation: PI Lab, Department of Computer Science and Technology

Sep 2025 – Jun 2028 (expected)

Shanghai Jiao Tong University

B.Eng. in Industrial Design · Rank 1/30 · National Scholarship

Semester exchange: University College London · Sep 2024 – Jan 2025

Jun 2021 – Jun 2025

INTERNSHIP EXPERIENCE

ByteDance | Seed

Feb – Jun 2026

LLM Data Product Intern | **LangGraph** Evaluation Agent

- **DataSeek post-evaluation analysis:** Developed and iterated a **Python/LangGraph** prototype that turned model-evaluation results into capability-level failure diagnoses and training-data recommendations. Led module requirements and analysis design through iterative releases; provided templates for four core analysis operators for engineering integration with Dataverse and R&D Agent orchestration.
- **Analysis pipeline design:** Designed an agent pipeline: reference reasoning → comparative error diagnosis → capability aggregation → training-data mapping. Used **best-of-N** reference sampling and LLM judging to establish a reasoning baseline; compared model and reference reasoning to extract question-level failure causes and capability labels. Aggregated fine-grained findings into mid-level capabilities for pretraining, SFT, and RL data plans. Designed a **four-quadrant analysis framework** (reference answer present/absent × scoring rubric present/absent) for objective and subjective evaluations.
- **Prompt system design:** Led the design of **30+** structured prompt templates with scenario-action capability names and constraints against unsupported capability attribution. To retain long-tail failure causes and make results comparable across evaluation sets, designed two-stage aggregation: first bucket by 18 predefined capability labels, then adapt grouping within each bucket to the number of failure causes.
- **Product design and iteration:** Independently wrote PRDs and led review rounds. Led manual quality reviews of **700+** reports, identified nearly 100 **P0/P1/P2** issues, and established issue-to-requirement tracking and fix follow-up. Designed short-term frontend fallbacks and longer-term prompt changes for recurring report failures.

Sohu | Mint Lab

Oct 2025 – Jan 2026

AI Product Management Intern

- **Knowledge-graph content creation:** Built an AI-assisted content tool for creators facing information overload and fragmented ideas. Designed the workflow around how creators develop and organize ideas, turning them into editable, traceable topic graphs, outlines, and drafts.
- **Prompt architecture and optimization:** Led prompt design across association → structuring → linear overview → drafting. Used **Root/Hyponym** constraints to organize root topics and child concepts; combined chain-of-thought prompting with self-ranking and voting to address uncontrolled expansion, hallucinations, and inconsistent topic hierarchies.
- **Engineering implementation:** Independently developed Python end-to-end test scripts. Reworked the generation pipeline by splitting title and description generation into sequential model calls. Designed an Outline-to-Draft flow so creators could edit structure before expanding it into prose and trace generated content back to its outline.
- **Evaluation system:** Built a **six-dimension** quality rubric with stage-specific weights and a four-level evaluation set from recurring content topics. Compared model combinations under quality, latency, and cost constraints; used regression cases to measure prompt changes and check for regressions across iterations.

INTERNSHIP EXPERIENCE · CONTINUED

Tsinghua University | Future Laboratory

Jun – Sep 2024

Human-Computer Interaction Research Intern

- **VR interaction research:** Reviewed **30+** papers and reports on cutting, chopping, and tearing in VR. Used findings on weak haptic immersion to define test scenarios and build a wearable force-feedback prototype.
- **Force-feedback modeling:** Modeled the relationship between motor input angle and output resistance. Developed three **C#/Unity** task modules for cutting fruit and vegetables, chopping a wooden target, and tearing paper.
- **User testing and iteration:** Tested the prototype with **eight participants**. Inspected contact-event timing, physical resistance, and signal traces; investigated mismatched feedback and compensatory effort near the end of cutting motions to guide prototype revisions.

PUBLICATIONS

“In Sync with Me”: A Robotic Body-Doubling Probe for ADHD Programmers’ Deep Work

Chenyang Li, **Sizhang Xu**, Hancheng Ye, and Ziyu Huang
UbiComp Companion '26 (Poster), 2026 · Accepted / in press

[DOI](#)

SELECTED PROJECTS

Real-Time Sleep Research Platform

2026

Python · **FastAPI** · Vue/JavaScript · **WebSocket** · BLE

- Built a **Python/FastAPI** and Vue research platform connecting BLE sensors, audio inference, and intervention control. Implemented REST controls and **WebSocket** state updates for live signals, disconnected devices, stale readings, and cooldown status.
- Separated audio capture, inference, and BLE I/O with a **ring buffer**, worker thread, and **asyncio** loop. Implemented the intervention **state machine**, **JSONL** event logs, and background waveform compression and **NPZ** writes.

Alpha-TradingAgents | Multi-Agent Equity Research

Summer 2026

Python · **LangGraph** · akshare | github.com/xuxu8421/alpha-trading-agents

- Extended the open-source TradingAgents framework for A-share research with market, financial, news, and policy data adapters and **exponential-backoff** retries. Added policy-analysis and report-writer nodes to **LangGraph**, reusing the framework’s bull/bear debate and risk-review orchestration.
- Defined a structured metadata interface between research synthesis and PDF rendering. Built summary cards from these fields and generated a **four-page** report combining analyst evidence, debate outcomes, and risk review.

CAMPUS LEADERSHIP

SJTU Student Union (35th term) — Presidium Member

Coordinated liaison and communication across 34 schools.

School of Design Student Union, SJTU (6th term) — Executive President

Nanyang News Agency, SJTU — Presidium Member & Deputy Secretary-General

Managed content for the university’s official WeChat account and led planning for 80+ campus-culture features. Team results included 170,000+ new followers and 1.75M+ article reads.

SKILLS

Programming & tools: Python, C#, Figma, Cursor (AI-assisted development)

Engineering: JavaScript, Vue, **FastAPI**, **WebSocket**

English tests: TOEFL 108; CET-6 641; CET-4 657